

adopted to create a tangible increase in efficiency and cost optimisation.

Fleet telematics can offer the basics of track and trace of vehicles on the move, but IoT can build intelligence into the process. IoT can integrate all processes, from loading to monitoring parameters in transit and final delivery. IoT tracks the estimated time of arrival (ETA), unloading/waiting at the destination, driving behaviour, vehicle and cargo safety, end-to-end process automation and workforce management.

IoT, with ML and AI, is helping companies optimise space, route, and operations' planning. Adopting IoT helps track assets on the move and ensures compliance to standards such as AIS140, hours of service, etc. This, in turn, ensures safety and enables the company to have a clear record for compliance needs. The IoT data also helps

management teams to analyse vendor performance and make informed annual contract decisions based on insights for factors such as quality, efficiency, etc.

Q. Are you doing anything to focus on industrial IoT solutions? What are the steps required to strengthen it?

A. Companies now offer rich and highly developer-friendly IoT platforms to build industrial IoT solutions for manufacturing units as well as help operations teams use IoT applications to maintain factory premises proactively. The platform and solution cater to a wide array of industries that include the manufacturing sector, oil and gas, as well as medium and large scale organisations that need centralised and proactive

monitoring and management.

WebNMS' industrial IoT capabilities empower industries to seamlessly converge physical and digital aspects of their business on a unified platform that can collect data, monitor and manage assets, perform remote automation and provide data analytics for day-to-day activities as well as strategic operations. The company's partner ecosystem enables it to seamlessly connect with PLCs, CNC machines, and other equipment on the shop floor, thereby enabling retrofitting IoT to existing infrastructure. The platform and solutions are aimed at helping manufacturers proactively monitor assets and processes, perform predictive maintenance, iron out process bottlenecks, and ensure the quality of production, all remotely and seamlessly. **END** 

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So, what advice would Pathak like to give to someone who wants to start out in the open source space? He says, "If you are a fresher, working on open source projects certainly gives you an edge over others to showcase your skills. Most importantly, you get to learn how the core of any system is built upon. You may not be a part of a big earning club, but you will be a part of the roots of the system. Even a small contribution to open source gives you a feeling of ownership and being a core member of that project."

And what are the must-have skillsets to make a successful career in FOSS? "The most important skill is being a team player. Also, contributions do not come only from developers; you can contribute to FOSS projects by translating, promoting, reporting bugs and features, etc," says Pathak.

He adds, "Since open source is not meant to generate profits, patience is the key. If you are looking to start an open source business, bootstrapping and building a community around your project will be the right way ahead."

The Indian open source space is still evolving

Pathak feels that the Indian open source space is still in the evolving stage. Open source is mainly about building tools and applications for the community. Since making things available for free is the main motto, you can't show a balance sheet to any investor to attract funding. This may not seem like an attractive business model to work upon, but is highly appreciated in other countries. It's because of this reason also that open source does not get so much recognition from the startup and media community as well.

Pathak says, "India ranks third when we talk of the number of contributions on GitHub, but there are few open source projects coming from this country. There are many communities building and releasing packages like CSS and JS libraries, but the number of proper enterprise open source applications can be counted on the fingers".

He adds, "Companies like Google, Uber, Airbnb, Tesla, etc, release many applications or libraries that they have

developed for their own use as open source so that the general developer community can make use of these. This is something that is found lacking in India MNCs."

Open source is being encouraged by the Indian government

According to Pathak, in the past few years, the Indian government has been promoting open source as a very important aspect of digital India and also as the way forward for an AatmaNirbhar Bharat. The policy of adopting open source software in all e-governance projects can certainly help leverage economic and strategic benefits. There has also been a rise in open source workshops and conferences around the country that focus on open source projects developed in India.

The future of open source sure seems bright. Pathak says that more and more companies will soon start to see the benefits of FOSS and the new opportunities it creates. Open source is certainly leading the way in the development of big applications and fulfilling an organisation's operational needs as well. **END** 